

$du = \delta Q - \delta A = Tds - pdv$
энтропия - дифференциал теплоты

Первое начало. Теплоемкость. Адиабатический и изотермический процессы.

"Кто бы это жидкость, знает?"

1.1 Решение
Дано $V = const$
 $T_1 \rightarrow T_2$
 $\Delta U = ?$

$\Delta U = \frac{i}{2} R \Delta T = \frac{i}{2} \Delta(pV) = 0$ $U_1 = C_V m_1 T_1$
 $pV = \frac{m_1}{\mu} R T_1$ $pV = \frac{m_2}{\mu} R T_2$ $U_2 = C_V m_2 T_2$
 $m_1 T_1 = m_2 T_2$

$du = \delta Q - \delta A = \delta Q - pdv$

1.2
 $V_0 \rightarrow 2V_0$
 $T = const$

$\delta A = p dV = \int_{V_1}^{V_2} p dV = \int_{V_1}^{V_2} \frac{RT}{V} dV = RT \int_{V_1}^{V_2} \frac{dV}{V} = RT \ln \frac{V_2}{V_1} =$
 $= RT \ln 2$
 $pV = \nu RT$
 $\frac{\nu RT}{V}$

1.3
 $T = 20^\circ C$
 $\Delta T = 1K$

$C_s^2 = \left(\frac{\partial p}{\partial \rho} \right)_s$, $p = \rho \frac{RT}{M}$

$du = \left(\frac{\partial u}{\partial T} \right)_v dT + \left(\frac{\partial u}{\partial v} \right)_T dv = C_V dT$

$\begin{cases} du = -pdv = C_V dT & p = \frac{m}{V} \Rightarrow V = \frac{m}{\rho} \\ p = \frac{\rho RT}{M} & \parallel \rho m d\frac{1}{\rho} \end{cases}$

$C_V \frac{M}{R} d\frac{p}{\rho} = \rho m d\frac{1}{\rho} = \frac{\rho^{m(-1)} d\rho}{\rho^2} = C_V \frac{M}{R} \left[\frac{1}{\rho} d\rho - \frac{p}{\rho^2} d\rho \right]$

$\frac{C_V M}{R} \frac{d\rho}{\rho^2} = \frac{C_V M p}{R \rho^2} d\rho - \frac{\rho m d\rho}{\rho^2}$

$\frac{C_V M}{m} = \tilde{C}_V$

$\tilde{C}_V \rho d\rho = \tilde{C}_V \rho d\rho + R \rho d\rho = \rho d\rho C_p$

$\rho d\rho = \frac{C_p}{C_V} \rho dV$

$\frac{\rho}{\rho_0} = \left(\frac{V}{V_0} \right)^\gamma$

$C_s^2 = \left(\frac{\partial p}{\partial \rho} \right)_s = \frac{\partial}{\partial \rho} \left(\frac{\rho_0 \rho^\gamma}{\rho_0^\gamma} \right)$

$C_s^2 = \frac{\rho_0}{\rho_0^\gamma} \cdot \gamma \cdot \frac{\rho^{\gamma-1}}{\rho} = \frac{\rho_0}{\rho_0^\gamma} \cdot \gamma \cdot \frac{\rho^\gamma}{\rho}$

$C_s^2 = \frac{\rho}{\rho_0} \gamma = \gamma \frac{p}{\rho}$

$C_s^2 \delta C_s = \gamma \frac{p}{\rho} \delta T = C_s^2 \delta T$

1.40. $C = const$ $C = \frac{\delta Q}{\delta T}$
Дано
 $V: 4l \rightarrow 1l$
 $p: 1 \rightarrow 8$
 $T: T_0 \rightarrow T_1$
 $300K$

$du = \delta Q - p dV$
 $\frac{du}{dT} = \frac{\delta Q}{dT} - \frac{p dV}{dT} = C - p \frac{dV}{dT} = C_V$
 $pV = \nu RT$
 $\tilde{C} - \tilde{C}_V = \frac{RT}{V} \frac{dV}{dT}$
 $\gamma \left(\frac{\tilde{C} - \tilde{C}_V}{R} \right) \frac{dT}{T} = \frac{dV}{V} \Rightarrow \left(\frac{T}{T_0} \right)^\gamma = \left(\frac{V}{V_0} \right)^\gamma$
 $\gamma = -2$
 $\tilde{C} = \tilde{C}_V - 2R$

1.54

$K = \frac{p_0}{V} s^2$ $F = p_0 \cdot s = \frac{K V_0}{s}$
 $p_0 V_0 = R T_0$
 V_0, R, T_0
 $y=1$

$C = \frac{\delta Q}{dT} = C_V + \frac{p dV}{dT}$ $F = \Delta p \cdot s = K \cdot \Delta h$

$R = \frac{p dV}{dT} + \frac{V dp}{dT} = \frac{p dV}{dT} \left(1 + \frac{p_0 V}{p V_0} \right)$

$V \frac{dp}{dT} = \nu K \frac{\Delta h}{s dT} = K V \frac{\Delta h}{s dT} = \frac{p_0 s^2 \nu \Delta h}{V_0 s} = \frac{p_0 V_0 \nu}{V_0 p} \frac{dV}{dT}$

$\Delta p s^2 = K \Delta h \cdot s$
 $\Delta p = K \frac{\Delta V}{s^2} = \frac{p_0}{V_0} \Delta V$

$\frac{5R}{2}$ $\frac{5}{2}R$

$p_0 V_0 = p T_0$
 $pV = RT$
 $\delta Q_1 + \delta Q_2 = 0$